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A Comprehensive Review of Optimizing Cloud-Based IoT Systems in Low-Connectivity Environments

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Abstract

Cloud computing has an important role in supporting Internet of Things (IoT) devices because it offers scalable storage, analytics and data processing. While deploying cloud-based IoT solutions in remote or low-connectivity environments, there are a number of difficulties arises like insufficient infrastructure, unstable network availability and excessive energy usage. This study's objective is to explore and assess the existing methods for enhancing cloud-enabled IoT devices' performance and effectiveness in such environments. By using review-based methods this study has investigated the current literatures related to intelligent task scheduling, hybrid cloud-edge computing and cloud resource optimization frameworks. Also, it assesses techniques which are used to manage low-network problems and improve energy consumption. The outcome explains how hybrid cloud-edge architecture and intelligent scheduling strategies help to improve system performance greatly. Though there are many current methods, they are not practical in rural areas and fail to fulfil all energy constraints. These current studies can give useful insights and solutions but there are still limitations in the development of cost-effective, dependable and consistent cloud-based IoT devices for rural locations. For future work, to improve its overall performance and support long-term deployment, it should focus on implementing energy-effective, flexible and offline-capable systems.

Keywords: *Internet of Things (IoT), Cloud Computing, Hybrid Cloud-Edge Computing, Edge Computing, Low Connectivity, Intelligent Scheduling, Resource Optimization, Rural Environments, Energy Efficiency, Green Computing*

1. Introduction

Cloud technology has become one of the significant technologies for supporting Internet of Things (IoT). Because it can provide processing power, data analytics techniques and scalable storage. Everywhere the growth of IoT devices is growing rapidly, for example healthcare monitoring, environmental sensing and smart agriculture. Because of this rapid growth, there is an increasing need for effective resource management and data processing. As traditional cloud-based methods still depend on stable internet networks, this causes major restrictions when it is deployed in rural or low connected areas. Because of low limited internet connection, system performance and reliability gets affected by these challenges like limited infrastructure, high latency and inconsistent connectivity.

Much research has been done to address this problem. Researchers have introduced edge computing and hybrid cloud-edge technologies in order to solve this problem. Here data processing is distributed between centralized cloud servers and local edge systems. According to Gill & Kaur (2025) in comparison to cloud-based methods hybrid methods can decrease delay, improve system performance and increase scalability. Similarly, in Andriulo et al. (2024) comparative analyses highlighted that systems can be improved with edge computing by processing data closer to the source and it can reduce network dependency and improve efficiency.

Here another important research area is IoT systems resource optimization and task scheduling. For ensuring optimal performance effective resource allocation and intelligent task scheduling are must in areas with limited resources specially. According to Vijayasekaran & Duraipandian (2022), the use of advanced approaches like deep learning for resource scheduling can enhance the performance of the system and lessen the response time. In another study, it investigated the scheduling approaches and shows that for improving overall system performance task offloading and dynamic resource management plays the most important role (Goudarzi et al., 2022).

Hybrid cloud-edge systems can prove their ability to reduce latency and enhance accessibility by implementing practically in real-world applications (Sanagari, 2025). According to Dankolo et al. (2025), in improving system performance and effectiveness, optimization techniques like multi-objective resource allocation play a significant part in IoT environments. Furthermore, many studies have investigated the advanced scheduling approaches for cloud-edge collaboration. In research of Shan et al. (2025), they highlighted the potential of advanced scheduling approaches to improve system intelligence and execution of the operations.

Even with these developments, there are still limitations to how these approaches are working in areas with less resources and bad internet connectivity. Because of not having real-world validation many existing research paid attention to theoretical models or simulations. That's why the aim of this study is to review and analyze existing works on intelligent scheduling, optimization of resources and hybrid cloud-edge IoT systems. The primary objective of this study is to find strengths, restrictions and research gaps and finally provide potential directions in order to improve cloud-based IoT frameworks reliability, efficiency and dependability in rural areas.

2. Literature Review

Recent works on cloud-enabled IoT systems primary focus is to enhance performance of system by using resource optimization, hybrid frameworks and intelligent scheduling. Hybrid cloud-edge frameworks have become the most widely discussed method which helps to improve system performance in low-connectivity areas. Regarding hybrid cloud-edge frameworks, Gill & Kaur (2025) highlighted their capabilities to improve scalability, improve resource utilization and lessen delays. They also explained that computation distribution between cloud and edge systems can overcome the restrictions of centralized systems (Gill & Kaur, 2025). As the study only focused on theoretical models, it is limited to experimental validation in rural or low-connectivity areas which is making it difficult to examine real-world efficiencies. This limitation is evidence that it needs more work on practical implementation and evaluation of performance in limited areas.

For improving the efficiency of systems another main area of cloud-enabled IoT systems is resource scheduling. Vijayasekaran & Duraipandian (2022) build a hybrid deep learning method to improve scheduling task and resource allocation, and the outcomes show that response time and system efficiency have improved significantly compared to traditional methods. But it is not suitable and reliable for rural areas with limited resources because the model needs high computational power and large datasets. This suggests that more simple and flexible scheduling techniques are required for practical real-world applications.

Andriulo et al. (2024) done a comparative analysis on cloud, edge and hybrid computing models and the study explained how hybrid approaches provide the best solution between performance and efficiency especially in requiring low latency in IoT systems. As the research lacks comprehensive review of resource optimization and scheduling strategies, it is crucial for improving system performance in poor-connectivity areas. And this shows the requirement for integrated research that combines architecture with optimization techniques.

Sanagari (2025) explored practical implementation of hybrid architectures, the study shows that hybrid applications can reduce latency and improve reliability effectively in IoT devices. Though having these advantages, the deployment focuses on controlled environments. It does not address fully problems such as poor connectivity or energy limitations in rural areas. This demonstrates a difference between practical deployment conditions and theoretical design.

According to Dankolo et al. (2025) research, they explained the resource optimization allocation and explained how it can improve system efficiency and reduce operational costs significantly by using their proposed models which were evaluated through simulations. But this does not address the real-world practical limitations like poor network and limited energy supply. As a result, there is a gap in understanding how these optimization methods operate in rural areas.

Goudarzi et al. (2022) reviewed the task scheduling techniques and their impact on IoT devices comprehensively. According to Goudarzi et al. (2022), task offloading and dynamic scheduling are the most important parts for improving system performance. As the primary target of this study is to focus on general IoT environments, it does not highlight particular constraints like poor internet connectivity and rural implementation scenarios. It needs more context-particular scheduling plans.

Luo et al. (2021) provided a comprehensive overview on scheduling approaches, load balancing and computation offloading. According to Luo et al. (2021), for having high efficiency in IoT devices efficient scheduling is the most important thing. Though the study explained about its comprehensive coverage, there are still gaps in energy efficiency and rural implementation problems. These are the significant factors in real-world practical applications. So, there is need for more research in constrained environments.

In a study of Shan et al. (2025), they highlighted recent developments in cloud-edge collaboration and they also proposed an advanced scheduling approach to improve the system intelligence and execution of work. According to Shan et al. (2025), because of task completion and resource utilization the system was improved. As the proposed method is a bit complex, it will require a lot of computational resources. That's why it is less suitable for low-resource areas. This indicates that there is need for less complex and adaptive solutions for rural IoT systems.

3. Materials and Methods

Here in this study, it combines review-based research methodology combined with comparative and analytical evaluation in order to investigate the existing techniques for improving cloud-based IoT systems in poor-connectivity areas. This study's methods are designed to investigate the limitations like lack of real-world applicability, insufficient integration of optimization and scheduling methods and limited focus on rural areas identified in past studies.

3.1 Design of Research

Systematic literature review method is a method where academic papers, conferences proceedings and journal articles are collected and analyzed relevant to the topic (Yu, 2017). In this study we are going to use this method and the purpose of using this method is to provide a comprehensive and well-structured understanding of existing techniques related to hybrid cloud-edge IoT systems, optimization of cloud resources and intelligent scheduling.

As for using this method, the literature was selected for this study based on given criteria:

1. Research papers are from recent publications (particularly from 2021-2025)
2. Full text of the literature must be available and credible sources.

3. It must be relevant to cloud computing and IoT systems
4. It must be focused on hybrid architecture, resource optimization and scheduling.
   According to Zhang et al. (2010) systematic literature review method is commonly used for comparing models, identifying trends and highlighting research gaps in cloud and IoT research.

3.2 Data Collection and Sources

This study data contains secondary data which were collected from online academic databases and published research papers. The sources include:

- IEEE Xplore
- SpringerLink
- Science Direct
- arXiv (Cornell University archive)
- Multidisciplinary Digital Publishing Institute (MDPI)
- ResearchGate
- PubMed Central

These sources are used for getting reliable and up-to-date information about current research patterns and advancements of technology.

3.3 Research Selection Criteria

A comparative analysis approach was applied in this study to fulfill the objectives of it and to cover the gaps found in the literature. The criteria are given below on which the selected research papers were analyzed:

1. Architecture of system (cloud, edge and hybrid).
2. Optimization techniques of resources.
3. Scheduling approaches.
4. Indicators for performance evaluation (latency, energy efficiency and response time).
5. Adaption in poor connectivity areas.

By using this approach and criteria, total eight existing research were collected for detailed investigations which include hybrid architecture, optimization and scheduling.

3.4 Extraction and Analysis Process of Data

For extracting and analyzing the selected studies it follows four-step approach. In Figure. 1 diagram the process of extraction and analysis are shown.

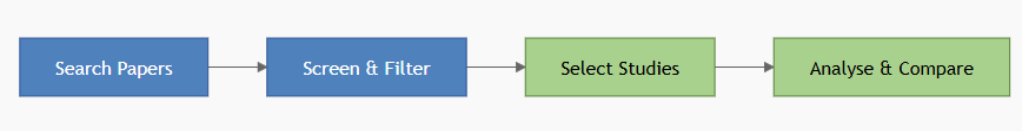


Figure 1. Flowchart for showing the selection of process this study used.

The brief explanations of these four steps are given below:

1. Identification : By using pre-defined keywords, relevant research papers were collected from selected databases.
2. Screening : To remove the irrelevant research papers related to the topic, title and abstract were reviewed thoroughly.
3. Eligibility : Based on the research selection criteria given, full text articles of every research paper were reviewed.
4. Analysis :

Based on the predefined evaluation indicators of performance, all the selected research papers were compared.

To make sure the study replicated and reduces selection bias, using this extraction and analysis process is must.

3.5 Evaluation Criteria

Without performance evaluating indicators here, we cannot compare and evaluate the selected research papers. That's why this study selected some of the specific performance evaluation indicators, those are given below:

1. Latency : This calculates the time which is required to process and respond back to tasks.
2. Scalability : It explains how much capability the system has to handle increasing workloads.
3. Resource Utilization : It shows how efficient the system is in using technological resources.
4. Reliability : It gives an idea about the system performance under limited or low network environments.
5. Energy Efficiency : This provides information about the system power usage during workloads.

In the given Figure 2 according to Goudarzi et al. (2022) the evaluation and scheduling perspective of edge and fog computing are shown.

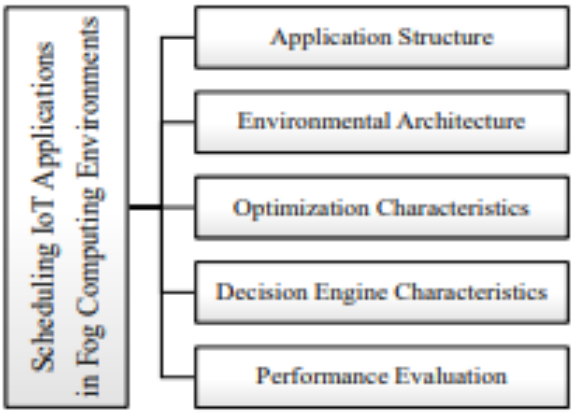


Figure 2. Evaluation and Scheduling Perspectives in Edge and Fog computing areas (Goudarzi et al., 2022).

According to Zhang et al. (2010), these performance indicators are frequently used in research of IoT and cloud computing for evaluating system performance.

3.6 Analyzed Models and Techniques:

This study is going to focus on four main techniques for analyzing. Those are discussed below:

- Architecture of Cloud-Edge Architecture

  According to Andriulo et al. (2024) by using hybrid system, it can distribute the processing tasks between the cloud devices and edge devices and as a result, it reduces the latency and can improve

the performance of the system in the low connectivity environments. According to Sharma et al. (2025) the architecture of hybrid cloud-edge systems is shown in Figure 3.

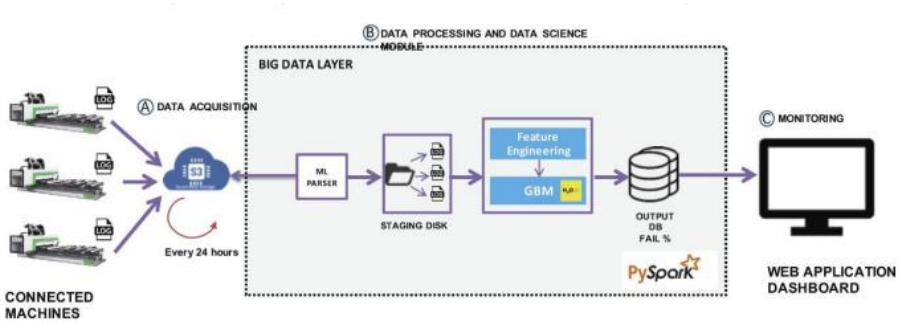


Figure 3. Architecture of Cloud-Edge hybrid IoT which shows the connection between devices, edge and cloud layers (Sharma et al., 2025).

- Optimization Techniques of Resources
  In a study of Dankolo et al. (2025), they proposed that techniques like load balancing and resource allocation dynamically can improve the performance of the IoT devices and it also reduces the operational cost for it.
- Scheduling Approaches Intelligently
  Vijayasekaran & Duraipandian (2022) suggested that AI-based task scheduling approaches play an important role in improving utilization of resources and it also helps to lessen the response time in cloud-edge environments.
- Frameworks for Advanced Scheduling
  According to Shan et al. (2025) advanced frameworks like cloud-edge task scheduling systems can improve the performance and coordination of the system, but the only issue is it will require more technological resources.

3.7 Used Tools and Platforms

In the selected research papers, those have used several platforms and tools for developing and simulating. It includes:

- Cloud Platforms : AWS, Microsoft Azure
- Simulation tools : MATLAB, CloudSim
- Programming Tools : Python

3.8 Research Gaps Identifying:

This study will discuss the gaps which were identified in the literature papers. For identifying the given methodology will be followed:

- For finding the lack validation of real-world applications, comparative analysis was applied in both theoretical and applied studies.
- By evaluating low-connectivity conditions it can prove the limited focus on rural areas.
- Combined analysis of optimization and scheduling approaches were used for showing the techniques were integrated weakly.

This study will provide more detailed and real-world understanding of cloud-based IoT devices by focusing on these research gaps.

To conclude, this study wanted to ensure a comprehensive evaluation of existing literature by using a structured and reliable methodology which combines systematic literature review and comparative analysis while identifying the main limitations.

4. Results and Discussion

4.1 Findings Overview

By reviewing the selected studies, the analysis highlights that cloud-based IoT devices have significantly developed with the use of edge computing and intelligent optimization methods. In the literatures, most of them proven that for better performance hybrid cloud-edge architecture works well particularly in terms of resource utilization, reduce of latency and scalability of system. In multiple studies these findings are identified which indicates that if computational tasks distribution is used between cloud and edge devices, then it will improve the systems performance efficiency in the areas with limited connectivity.

As the outcomes also shown that these techniques only perform well in simulated or controlled environments, that's why their effectiveness for real-world rural places is still not efficient. As a result, it proves that there is a gap between theoretical advancements and practical development.

4.2 Cloud, Edge and Hybrid Frameworks comparison

Based on the approaches, strengths, suitability and limitations **Table 1** shows the comparative analysis of selected studies for rural and low network area.

Table 1: Comparative analysis of cloud-enabled IoT frameworks.

Study	Approach Used	Key Contribution	Strengths	Limitations	Suitability for Rural / Low Connectivity
Andriulo et al. (2024)	Cloud vs Edge vs Hybrid Review	Compare architectures for IoT systems	Clear comparison of models, explains hybrid benefits	Limited focus on real-world rural deployment	Moderate – useful for understanding architecture but lacks rural focus
Dankolo et al. (2025)	Resource Optimization (Metaheuristic)	Improves resource allocation efficiency	Reduces cost and improves performance	Tested mainly in simulation	Moderate – effective but not validated in rural conditions
Vijayasekaran & Duraipandian (2022)	Deep Learning Scheduling	Optimize task scheduling in IoT networks	Improves response time and efficiency	Requires high computational power	Low – difficult to implement in low-resource rural areas
Gill & Kaur (2025)	Hybrid Edge–Cloud Review	Reviews of hybrid architectures	Strong overview of system designs	Mostly theoretical analysis	High – directly relevant to low-connectivity environments
Goudarzi et al. (2022)	Scheduling Taxonomy	Classifies scheduling methods	Comprehensive analysis of scheduling techniques	Lacks rural-specific focus	Moderate – useful but not context-specific
Luo et al. (2021)	Resource Scheduling Survey	Reviews of scheduling and offloading techniques	Covers wide range of techniques	Limited focus on energy and rural issues	Moderate
Sanagari (2025)	Hybrid Cloud–Edge Implementation	Real system design using cloud-edge	Practical implementation and architecture	Limited testing under real rural conditions	High – strong practical relevance
Shan et al. (2025)	Cloud–Edge Scheduling Framework	Advanced workflow scheduling system	Improves system coordination and performance	Complex and resource-intensive	Low–Moderate
Sharma et al. (2025)	Hybrid IoT Architecture	Proposes integrated cloud-edge model	Simple and efficient architecture design	Limited evaluation	High – suitable for rural deployment

By comparing different computing methods, it shows that each method has its own unique advantages and limitations. Its advantages and limitations are discussed below:

- Only cloud devices provide the most computational power and reliability. But it is highly dependent on stable internet connection which makes it less suitable for rural areas.
- Only edge systems can help to reduce latency and dependency on network by processing data locally. But it is restricted by low computational capacity.
- Hybrid cloud-edge systems combine the strengths of both methods by using cloud resources for complex tasks while enabling local processing at the edge.

The outcome proves that for low-connectivity areas hybrid frameworks are the best solution as they can balance performance, scalability and efficiency. The problem is it adds complexity because of its system design and requires advanced management approaches.

4.3 Resource Optimization Techniques Effectiveness

Dynamic resource allocation and load balancing are the techniques of resource optimization. It plays an important part in improving system performance. From the findings it indicates that it helps to reduce computational delays, improve utilization of resources and it minimizes the operational costs. For considering multiple indicators like latency, energy consumption and task distribution simultaneously, multi-objective optimization method will be the most effective but this may not perform well equally in real-world applications with unstable internet connection and limited infrastructure.

4.4 Intelligent Scheduling Impacts

AI-based and machine-learning methods for scheduling are known as Intelligent scheduling methods. These methods have shown improvements in task management and system efficiency significantly. The outcome shows that it can reduce response time, improve system throughput and enhance the utilization of resources. Though having these advantages, it needs high computational power and continuous input data which is not suitable for rural areas. And because of this, it makes a limitation in their real-world applications with low-resource settings.

4.5 Low-Connection Rural Areas Problems

This study finds many primary problems in implementing cloud-based IoT devices in rural areas, those are:

- Unreliable network connection means when data transmission gets disrupted.
- Limited infrastructure which means lack of internet and computing resources.
- High energy consumption when continuous data is processing.
- Highly prices when making advanced systems which are difficult to implement.

These four issues affect the performance and reliability of IoT devices significantly. And suggests the requirement for more flexible and energy efficient solutions.

4.6 Practical Applications and Significance

For practical rural scenarios cloud-based IoT devices have become one of the essential things, for example:

- As for precision agriculture, they use sensors to monitor the conditions of soil and help to optimize irrigation.
- In healthcare, they are using remotes for monitoring patients and telemedicine.
- For protecting environment, they monitor air quality and detecting disaster.

These outcomes suggest that effective scheduling and optimized resource management can upgrade the applications performance significantly by decreasing the delays, conserving energy and making sure operations are continuous even in low internet connectivity area.

4.7 Research Gaps Discussion

Though having significant advantages in cloud-based IoT systems, there are still several research gaps which restrict their real-world effectiveness in areas like with limited resource or low-internet network. One of the major issues is the lack of practical real-world applications and testing. Many existing research depends on simulations or controlled areas, but this dose not capture real-world issues like hardware limitations, unstable network conditions and environmental variability. That's why the proposed methods of performance remain uncertain in rural environments.

Then next major gap is the less focus on efficiency of energy and sustainable solutions. In the studies some considered performance improvements, but there is less number which address too much energy consumption connected to continuous processing of data and communication which become an issue in rural areas. Because ere energy resources are often few and unreliable there.

Another limitation of this topic is that implantation of resources optimization and intelligent scheduling approaches are not sufficient. Most of the research papers investigate these factors separately which lessens the effectiveness of the system. A more developed method could make system more efficient and reliable significantly.

Lastly, there are less standardized methods for implementing cloud-based IoT devices in rural areas which create difference in device design and evaluation. It makes it more difficult to compare solutions or ensure reliability without common guidelines. As a result, these also suggest the requirement for more real-world, developed and sustainable research methods.

4.8 Future Guidelines and Implications

This study's findings will provide essential information which will help future research to improve the performance and effectiveness of cloud-based IoT devices in poor-internet connectivity areas. First and main direction is the implementation of flexible and adaptive scheduling methods so that it can efficiently process with less computational resources (Goudarzi et al., 2022) (Luo et al., 2021). Because such methods have the ability of adjusting to changing network conditions and demands of tasks dynamically. This makes it perfect for rural implementation.

Second most important part is the improvement of energy efficiency computing techniques. For future framework, it should be developed with low-power devices, energy-efficient scheduling and renewable energy sources so that it can reduce energy consumption overall. For ensuring a long-term sustainability system this is essential in limited resource areas (Dankolo et al., 2025) (Gill & Kaur, 2025). Furthermore, based on the gaps there is also need for design system which will support low-network or offline processes. For this kind of scenario, techniques like caching, local data processing and delay tolerant communication can work well to maintain the framework functionality (Andriulo et al., 2024).

Lastly, as many research papers depend on simulations rather than real deployment in rural environments, future work should be focused on real-world testing and validation. Because real-world implementation

will provide better and accurate insights into system performance and usability (Sanagari, 2025) (Shan et al., 2025).

These given directions can improve the reliability, sustainability and scalability of cloud-based IoT devices significantly overall.

5. Conclusion

This study reviewed the existing research on cloud-based IoT optimized frameworks in low-internet network areas. Its focus was architecture of hybrid cloud-edge, optimization of resources and scheduling techniques intelligently. The finding of this study suggests that hybrid cloud-edge frameworks are more efficient than traditional cloud frameworks. Because they reduce delays, improve the utilization of resources and improve the system's reliability. Furthermore, for improving performance, optimization and scheduling approaches play a important part, but still their effectiveness is limited most of the time by high computational need and lack of real-world validation.

It also highlights multiple challenges which includes unstable connectivity, limited infrastructure, and too much energy consumption in rural areas. These challenges reduce the practical acceptance of many existing framework solutions. That's why building a lightweight, flexible and energy-efficient framework has become essential for making the performance better in such conditions.

As per terms of future direction, more research should target real-world deployment, computing energy efficient and the better development of optimization and scheduling methods. These improvements can be seen in areas like precision agriculture, healthcare monitoring and environmental monitoring potentially. To conclude, this study highlights the importance of building reliable, scalable and sustainable cloud-based IoT frameworks for encouraging the development of new technologies in the low-network connectivity areas.

Declarations

This study is not funded research, it did not receive any external funding. In this study the research papers that have been used are collected from publicly available academic sources and databases. All relevant references have been cited properly within the document. As it does not involve human or animal subjects and it is only a study based on existing literature review, ethical approval was not required. The author of this study declares that there is no conflict of interest related to this study.

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